Intelligent Information Systems Part 2

rz20405

April 20, 2024

1 Introduction to the scenario

The multi-agent system created in Jason is a restaurant scenario, with a client (the customer), a host and a floor manager. The customer enters the restaurant and requests a table. The host greets the customer and notifies the manager. The manager gives the host the go-ahead to find a table. The host then asks the manager whether there is a table available, and the manager communicates this back to the host. The host either assigns the customer a table, or tells them to try again. The client is set on a loop; after leaving the restaurant, they return after 1500ms, and if they do not get a table, they will come back to try again.

2 Implementation

There are three agents, with separate .asl files. Each agent is also instantiated in the .mas2j file. The agents communicate via the .send function to update each other and trigger an event.

The client has the initial goal to !get(table), and the floor_manager has the initial belief for the number of available tables. The manager also has the rule IF free_tables(X) AND X > 0 THEN table_available. The belief is updated when the client sits at a table and when the client leaves.

The first variation is implemented if the initial belief for the manager means that tables are available, i.e. free_tables(10). In this case, when the host asks the manager if there is a free table, the manager checks availability. This should meet the rule that the number of tables is greater than 0, and so the manager tells the host to assign a table.

The second variation is implemented if free_tables(0), i.e. there are no free tables. In this case, the manager reaches a different conclusion due to the negation **not** tableavailable and the host therefore has to tell the client that no table is available.

In each case, the scenario repeats itself as the client waits before returning to the restaurant in an endless loop.